INTRODUCTION

INTRODUCTION

A. System References

Online payment allows for cashless payments for goods and services through cards, mobile phones, or the internet. The main advantages of using online payment are cost and time savings, increased sales, and reduced transaction costs.

Online payment gateway provides systemized payment methods. It facilitates the customer with a fast and flexible way to pay.

Online Payment Gateway provides many options for transaction.

One of the most important features to consider in a payment gateway is security.

A payment gateway is the front-end technology that reads payment cards and sends customer information to the merchant acquiring bank for processing.

A payment gateway validates the customer's card details securely, ensures the funds are available and eventually enables merchants to get paid. It acts as an interface between a merchant's website and its acquirer.

In order to streamline the process of encryption, decryption and how payment integration actually works, we have developed Web-based application in order to demonstrate the payment signature with proper security. This was developed and designed to present how an Online transaction works and how we can verify whether transaction is fraud or not.

An assistive chatbot interface is also provided for the users to know the information about this website and to solve the queries of the users. This chatbot is implemented using Diaglogflow and that model is being trained for real-time users.

B. Overall Description:

What is Payment Gateway:

A payment gateway as a merchant service that processes credit card payments for ecommerce sites and traditional brick and mortar stores. Popular payment gateways include PayPal/Braintree, Stripe, and Razorpay.

Steps to implement Payment Gateway:

1. Encryption: For Security reasons, the data fetched from user should be encrypted into cipher text to ensure the privacy of the user
2. Request: The authorization request occurs when a payment processor gets approval from a credit card company or financial institution to proceed with the transaction.

3. Fulfilment:
   When the payment gateway has the authorization, it allows the website and interface to proceed to the next action.

Razorpay provides payment-gateway services used by more than 30,000 merchants. Razorpay is the only converged payments solution company in India that allows your business to accept, process, and disburse payments via its product suite. With Razorpay, you have access to all payment modes, including credit and debit cards, UPI, and popular mobile wallets.

Razorpay provides two keys for integration; private key and public key for integration.

What is Django?

Django is Web Framework, which follows MVT (Model View Template) architecture.

Django is a high-level Python web framework that enables rapid development of secure and maintainable websites. Built by experienced developers, Django takes care of much of the hassle of web development, so you can focus on writing your app without needing to reinvent the wheel.

Django is an open-source framework for backend web applications based on Python.

This framework provides the front-end and back-end and built-in database (SQLite). Also, Django supports multilanguage such as HTML, JavaScript for UI designing.

What is Razorpay API?

Razorpay APIs are completely RESTful and all our responses are returned in JSON. You can use Razorpay APIs in two modes, **Test** and **Live**. The API key is different for each mode. Know about generating API Keys.

Key ID and API Key are used as security credentials for authenticating the user. They are used to prevent your account from being abused to get your Key ID and API Key, log in to your Razorpay account and verify that you're in Live Mode.'

Razorpay provides API keys which should be integrated in project in order to use the facilities of Razorpay.

Also, checkout and payment signatures are to be implemented in order to verify that the payment done is successful.

What Django and Razorpay Integration do for us:

Integrating Razorpay to Django provides a payment gateway for various transactions such as for donation marketing, stores, etc.

- Provides proper authentication for login before using this website.
- Sends authentication tokens and links for resetting password and changing password.
- Provides email alerts whenever transaction is succeeded.
- Provides real-time chatbot for solving the queries of user.
- Provides various payment methods.
- Provides payment signature and capture.
- Provides appropriate security policy.
- Provides acknowledgement letter.
- Provides feedback system.

C. Software Project Constraints

An Application Programming Interface (API) is an interface that facilitates communication and connectivity between two or more software applications.
API Integration refers to this seamless connectivity that allows organizations to automate business processes, and enhance the sharing and embedding of data between various applications and systems.

API integration is the connection between two or more applications via their APIs (application programming interfaces) that allow systems to exchange data sources. API integrations power processes throughout many sectors and layers of an organization to keep data in sync, enhance productivity and drive revenue.

API is an interface and is part and parcel of almost everything in the digital world. No matter the business and the size of the enterprise, APIs enable seamless operation and performance of applications and web systems.

Benefits of API Integration:

1. Automation
2. Scalability
3. Streamlined Visibility
4. Increases productivity
5. Enhances Customer Experiences

PROJECT SCHEDULE

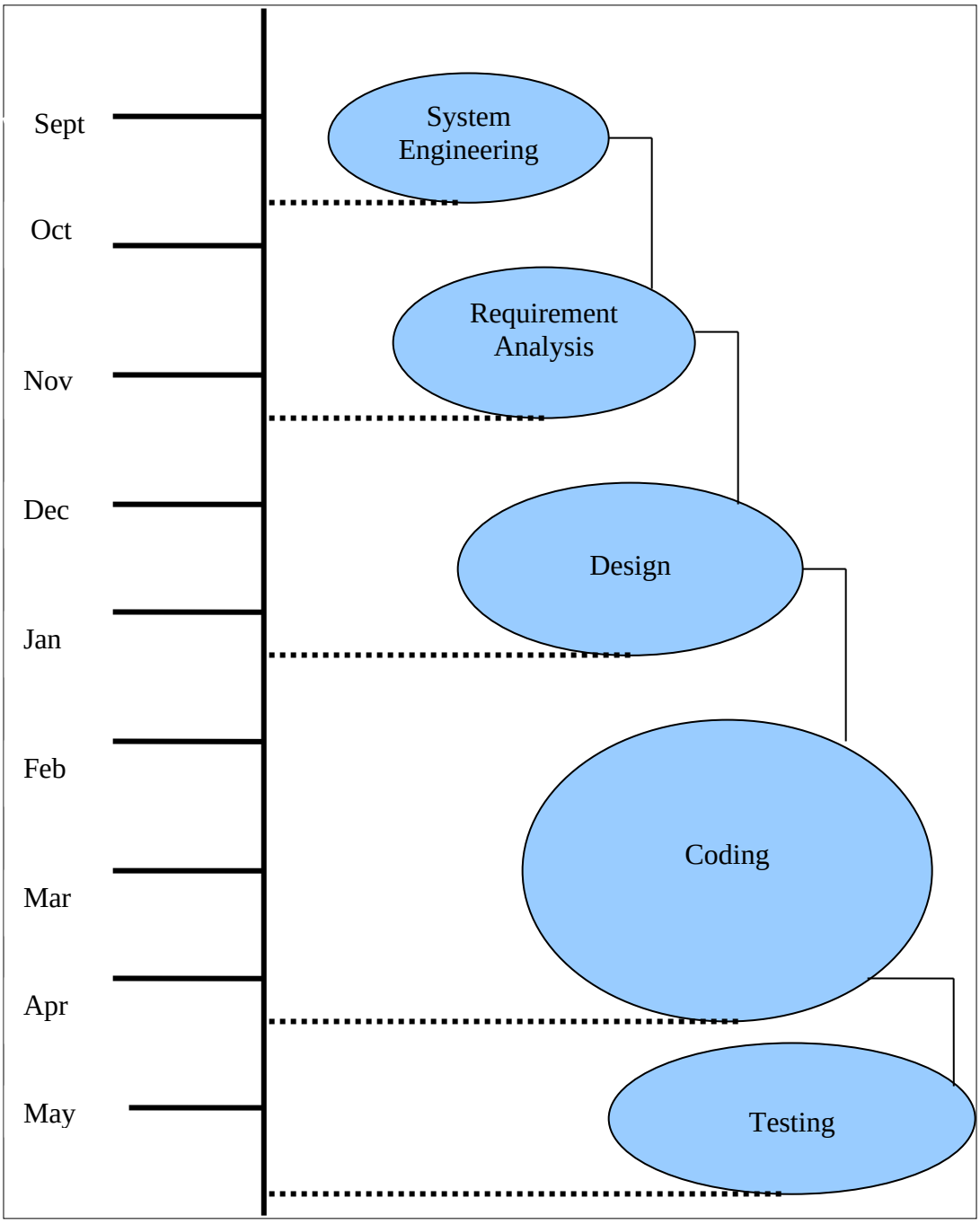
PROJECT SCHEDULE

Software Engineering is an establishment of computer engineering principles in order to obtain economical software that is reliable and works efficiently on real machine.

We have used classic life cycle paradigm also called “Water fall model”. For software engineering which is sequential approach to software development that begins at the system level and progress through analysis, design, coding, testing and maintenance. We had completed system engineering and analysis by the end of September 2021 which encompasses requirement gathering at the system level in the small amount of top-level design analysis.

We had completed software requirement analysis by the mid of October 2021 which encompasses both system and software requirement gathering. By the end of October 2021, we had completed the project planning and design. On the basis of design prepared in the previous stage by the end of April 2022 we completed coding stage.

After completion of coding page, the important part of the software development is carried out, which is the testing phase by the mid of May 2022. Various criteria of testing were taken in account has unit testing, integration testing, validation testing and system testing. First, each and every module of the project was tested under the unit testing. After unit testing, integration testing was carried out by integrating all module tested in unit testing. After unit testing the module prepared was cross checked with the design. At last, the whole module was tested over the system and desired result was obtained.


Sept
Oct
Nov
Dec
Jan
Feb
Mar
Apr
May
System Engineering
Requirement Analysis
Design
Coding
Testing

REQUIREMENTS ANALYSIS

REQUIREMENTS ANALYSIS

Software Requirements Specification:

1. Problem Statement -

To Integrate Payment Gateway to a Website By using Razorpay.

2. Objectives

This web application consists Super User and customers. Super User controls all encryption activities and data handling at database. However, the major objectives are:

- Login/ Signup using proper authentication and authorization
- To demonstrate Payment Integration using Razorpay.
- End-to-End Transaction

Software Requirements

1. Operating System Support-

There must be used are windows-Xp / Mac operating system and android (for application) which will support our functions that we are going to build in our project.

2. Programming Language -

This project is mainly based of Desktop based so, we choose Python-Django for backend and HTML, CSS, JavaScript for Front End UI.

3. Constraints-

The concept of Cashless Transaction significantly increases over the time and the effectiveness of an information of a system spread among the users. To use it effectively. It encompasses a broad set of methods and tools related to understanding and modifying system information.

This application is basically a Desktop application system targeted for society from where many people can access and perform online transactions of verify their signature.

The main constraints of our project are the user can easily access and performed the operations. In this way user can maintain the end-to-end reliability using the internet.

3.1 Performance Analysis

Whenever user log in into the website and perform any transaction based on the state of transaction, user is provided the email of the status.

3.2 Technical Analysis

As the project is based on desktop application Python-Django is used for backend language and HTML, CSS, JavaScript is used for front-end and SQLite is used for storing the data.

As Python is easy to use, supports multi-platform development and multi-paradigm, we can also parse data from one language to another. Django is a framework of Python which provides excellent security
And built-in database.

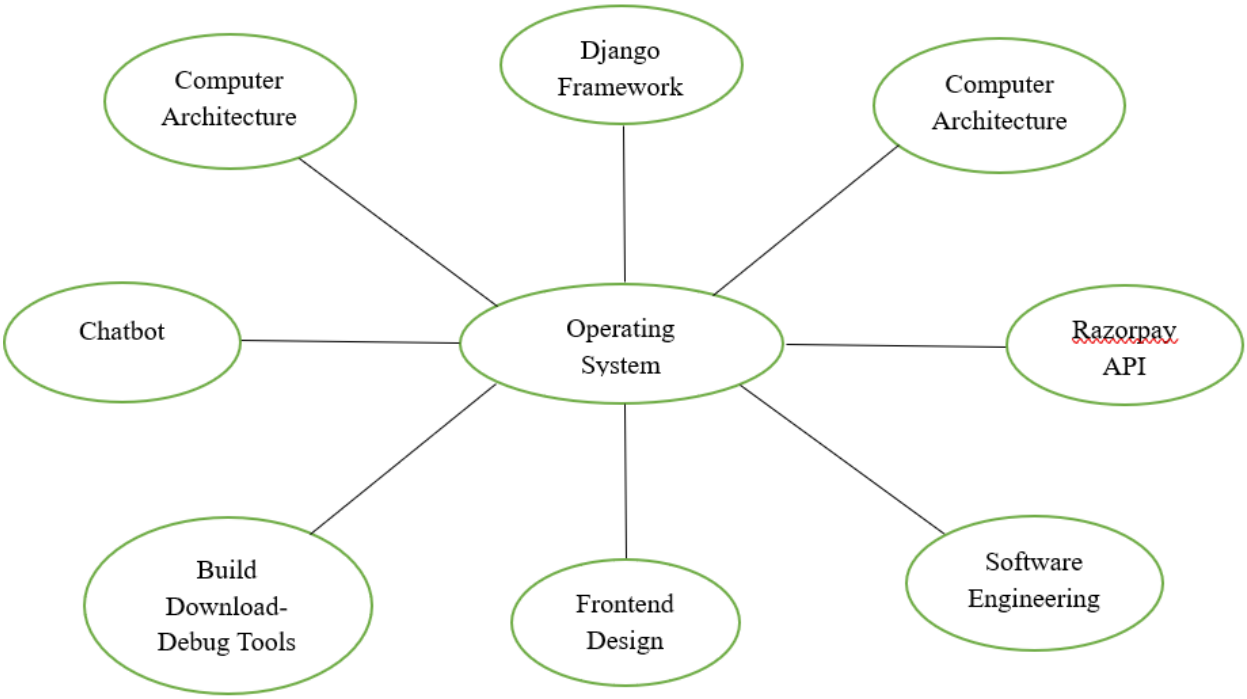
Dialogflow is a natural language understanding platform that makes it easy to design and integrate a conversational user interface into your mobile app, web application, device, bot, interactive voice response system.

Training the bot from intents and responses, this chatbot is being implemented to solve the queries of the user.

DESIGN

Analysis Modeling –

System Design Calls:



Working Design Cycle:

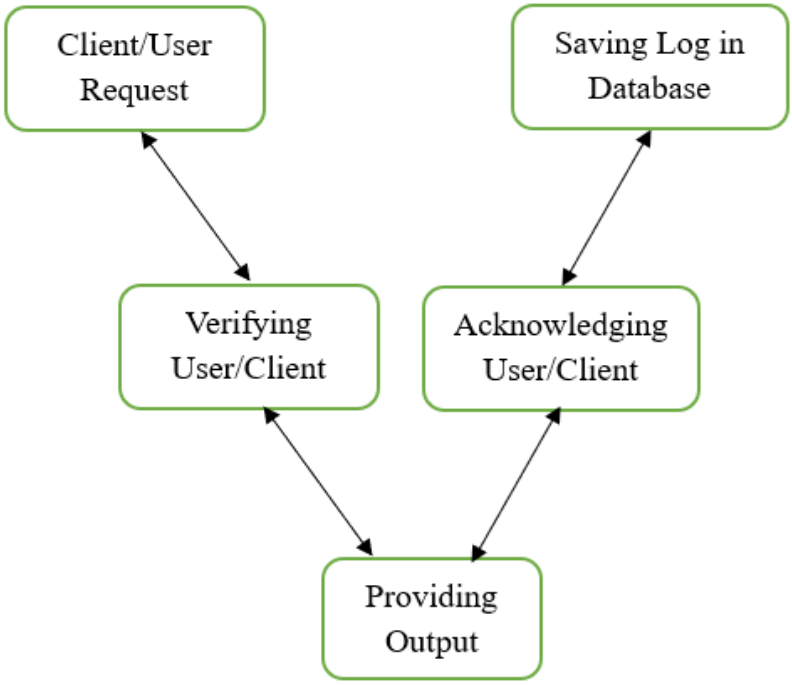


Fig 1.2

Characteristics of Django Framework

- **Excellent Documentation**

  If we compare Django with other open-source technologies, it offers the best documentation in the market.

- **Python Web-framework**

  Python is also one of the main reasons people started learning Django. It is that one tool which can solve all your problems and in any kind of operation out there, we can use it. It's very simple and easy to use.

- **SEO Optimised**

  This is a special feature of Django due to which it has edge over others. SEO is Search Engine Optimization as from the name it means that adding your website to the search engine such that it appears in the top results.

- **High Scalability**

  A lot of MNCs on a worldwide scale uses Django and it gets implement there without any defects or errors. It is the best example of Django being scalable.

- **Versatile in Nature**

  Django is very versatile in its own Django way. The logical project structure and MVT architecture of Django sometimes seem very limiting. But, that's just opposite because by giving us the files it is providing us with a solid foundation which can then be used to make whichever application we want to create.

➢ Functional Partitioning:

1. Login/Signup
2. Type Selection
3. Verification of data
4. Status of Payment

➢ Functional Description:

1) Login/Signup

Input:

Login using existing account/create new account

Output:

Log in successful/ account created successfully.

Specification:

This module provides the functionality to create an account or login into the website using existing account. If account is not created then, the user cannot access the main website.

2) Name: Type Selection

Input:

Select Available types in Dropdown List

Output:

Display the selected type in Dropdown list

Specification:

This module provides to select a specific type using dropdown box. In this case which helps user to select which type of Donation they wish to do for example Donation for children and donation for their expenses.

3) Name: Verification of Data

Input:

Type selected, amount wish to donate and email id to send acknowledgement

Output:

Verified data which is either correct or incorrect.

Specification:

This module verifies the data i.e., user has given valid email id, amount, type of donation and most importantly payment details given by user.

4) Name: Status of Payment

Input:

Boolean value which is obtained after verification of data.

Output:

Payment is successful or failed.

Specification:

This module checks the Boolean value obtained and decides whether the payment is completed successfully or unsuccessfully

➢ BEHAVIORAL MODELING

Block Diagram:

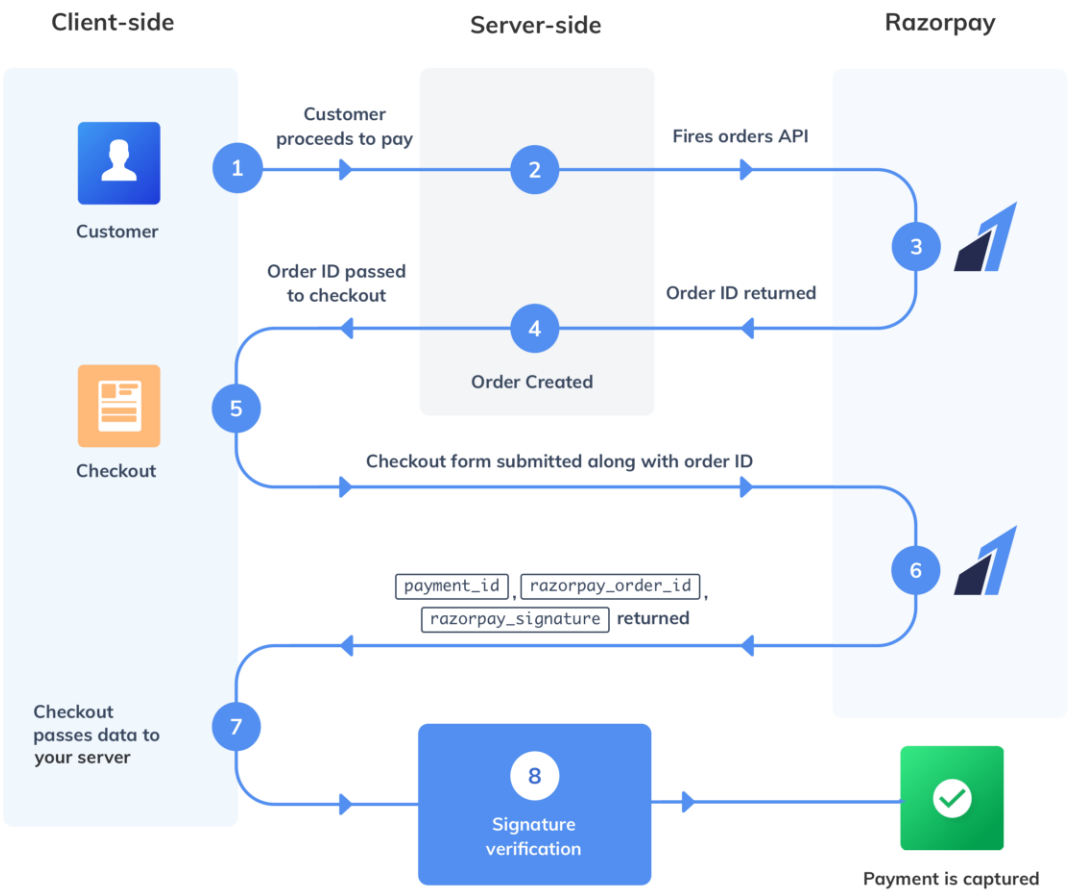


fig1.3

Deployment Diagram-

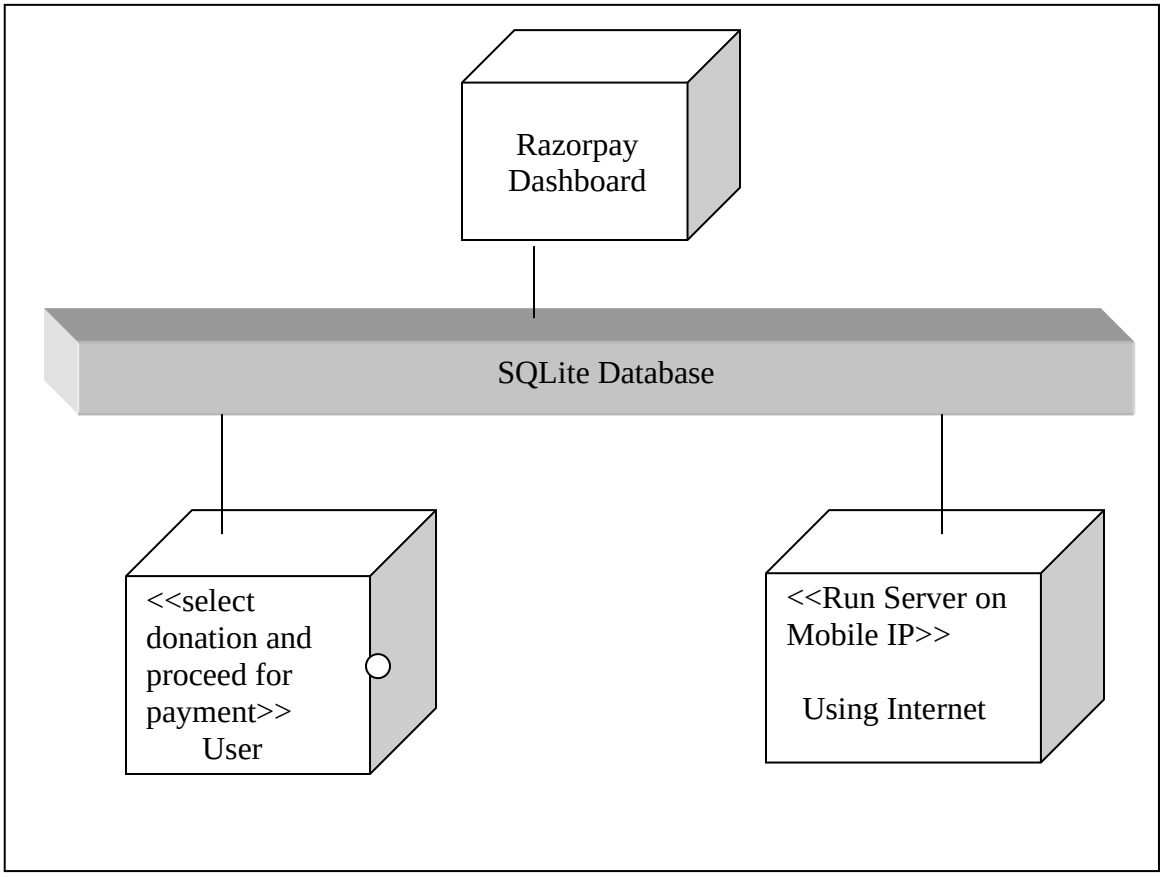


Fig1.4

IMPLEMENTATION

IMPLEMENTATION

➢ Proposed Architecture:

Language:

This project is based on the Desktop application, so we choose Python-Django, JavaScript and Bootstrap to develop our project.

Tools used:

- Google cloud (Dialog flow)

Software requirements

- Visual Studio Code- JavaScript, Python-Django
- SQLite
- Client OS - Microsoft Windows XP, android.

➢ Overall Description

Django is a high-level Python web framework that encourages rapid development and clean, pragmatic design. Built by experienced developers, it takes care of much of the hassle of web development, so you can focus on writing your app without needing to reinvent the wheel. It's free and open source.

Integrate Razorpay Standard Checkout with your website to start accepting online payments from your customers. Razorpay supports a various modes of payment methods such as net banking, credit and debit cards, wallets and UPI. Our Standard Checkout library provides all the essential features for integrating Razorpay Checkout with the client-side of your application. This is available only for web-based integrations.

For this purpose, a special framework of python "Django" is used. As Django follows the principle of "Model-View-Template" architecture, it is possible to demonstrate the three-layer architecture in payment gateway integration.

The main focus is to use most of the open-source data efficiently. Razorpay is an open source and easy interface for processing online transactions. It provides standard checkouts for your website to start accepting online payments from customers. Razorpay supports various payments such as UPI, Net Banking, Credit and Debit cards, QR code.

Initially, user need to create an account to access this website or to login with their account with proper credentials. If the user is new, then he needs to verify his account by the token link sent to their respective email.

After that user will be provided the website once the mail is verified. On the website in order to demonstrate how actually online transactions works using third party interface, we have provided sample products with their information and price and depending on that price the transaction will be carried out.

After selecting the product and providing data to the Razorpay API, a unique key will be generated that will be the response of API in form of JSON. Another key will be generated at the backend and stored in database.

The key generated and the key returned by the API will be compared for security purpose and based on that value further activity will be decided. After that payment methods will be provided for the user to complete the transaction. One Time Password will be sent to their respective mobile number for verification purpose.

As part of the online transaction, how actually the transaction is carried out, the demo of the online payment is implemented using Razorpay. Also, a navigation bar is also provided which includes the information about the project, members and the feedback option.

The Chatbot is also integrated with this website using Diaglogflow which solves the queries of real time users.

The basic process includes handling user data, processing data and providing the resulted output are organized differently in different development process. The team is will be divided to work on each module of project. All the members of our team will play separate roles in each part of the projects and roles assigned to them is carried out accordingly.

TESTING REPORT

TESTING REPORTS

1. Unit Testing

Unit testing focuses each module individually, ensuring that if function properly as unit. In this testing we have tested of our software to ensure maximum error detection. It helps to remove bugs from sub modules & prevent arrival of huge bugs after words.

Functional Partitioning:

- Login/Signup before accessing the website
- Integrating the API keys of Razorpay.
- Verifying the Payment Signature.

Functional Description:

1) Login/Signup: Restricts the access of unauthorized users

Input:

Username and password

Output:

Message displayed on screen based on the validation.

Specification:

This module restricts the access of the users if they do not have an account for this website.

2) Integration: Connecting Razorpay Dashboard using secret and public API keys

Input:

API Keys

Output: Connection Established.

Specification:

This module establishes a connection of Razorpay Dashboard and localhost's website.

3) Payment Signature: After every Transaction, payment key and RazorPay key are compared for payment capture.

Input: Payment Id Fetched from Database

Output: Captured/ Created/Cancelled based on the comparison result

Specification:

This module checks whether the payment is captured or not.

2.Integration Testing:

Integration testing is a systematic technique for constructing the program structure while at the same time conducting tests to uncover errors associated with interfacing. The objective is to take unit tested components and build a program structure that has been dictated by the design.

We prefer the Top-down integration testing as a testing approach for our project. The Top-down integration testing is an incremental approach to construction of program structure. Modules are integrated by moving downward through the control hierarchy, beginning with the main control module

Modules subordinate to the main module are incorporated into the structure in depth-first order.

Depth-first integration would integrate all components on a major control path of a structure. Selection of a major path is somewhat arbitrary and depends on application-specific characteristics.

3. Validation testing

The purpose of validation testing is to ensure that all expectations of the customer have been satisfied. The testing is done by the project group members by inspecting the requirements and validating each requirement.

We prefer the alpha testing technique for the validation purpose. In which case the group of students is called on to test the services of Razorpay. The group members will present at the place and observing the working and collecting the bugs

occurring at the site. The changes are made according to the requirements and testing is done again to gather more errors if present

4. System Testing

In the system testing system undergoes various exercises to fulfill the system requirements. These tests include:

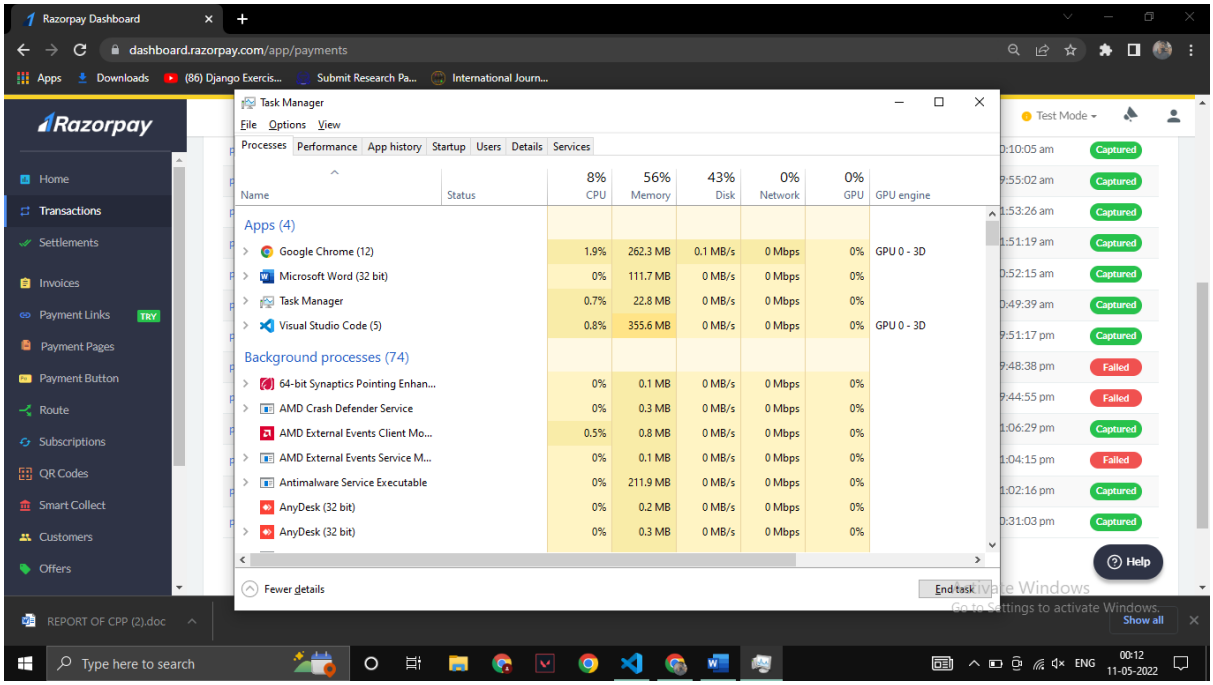
a. Security Tests: these are designed to ensure no user can access the other documents which are none of his business,

b. Performance Tests: The tests are conducted to check the performance of the system.

PERFORMANCE ANALYSIS

PERFORMANCE ANALYSIS

➢ Figure below shows the details of performance analysis of Razorpay.com



INSTALLATION GUIDE

INSTALLATION GUIDE

1. Operating System: Microsoft windows or android
2. Python interpreter needed
3. Virtual Environment for dependencies.

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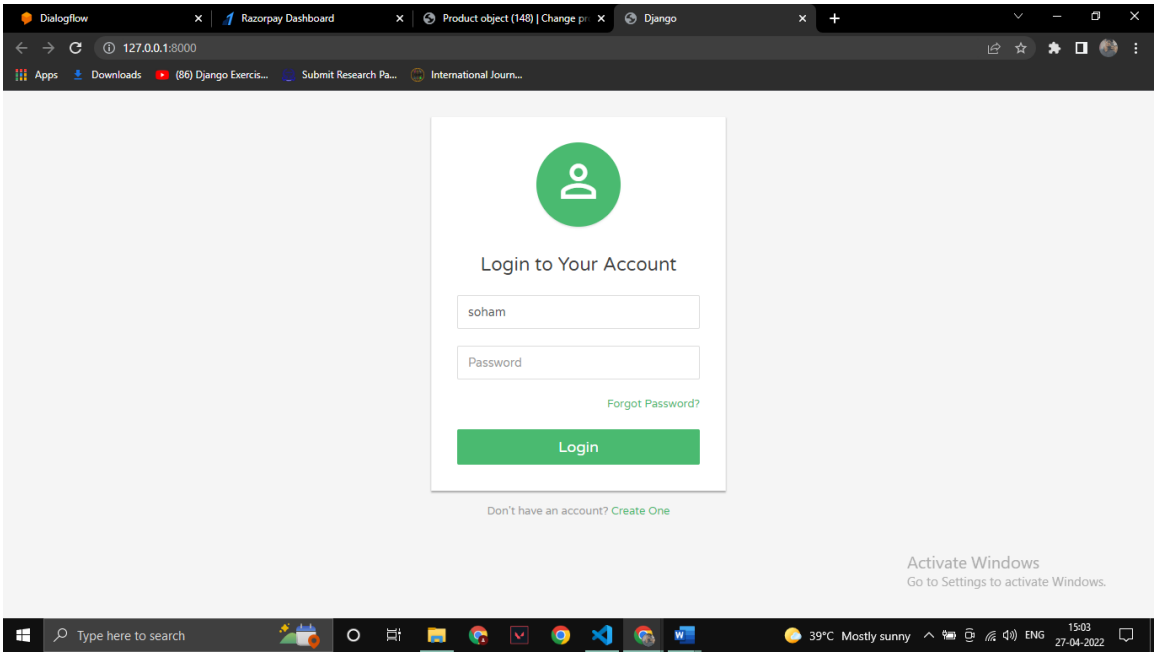
Install Procedure for Client

- Browse for www.razorpay.com, enter the valid username and password and login into dashboard.
- Hit Run server Command and Open Localhost.
- Login using username and password or create new account if required.
- Choose your donation and proceed for checkout.

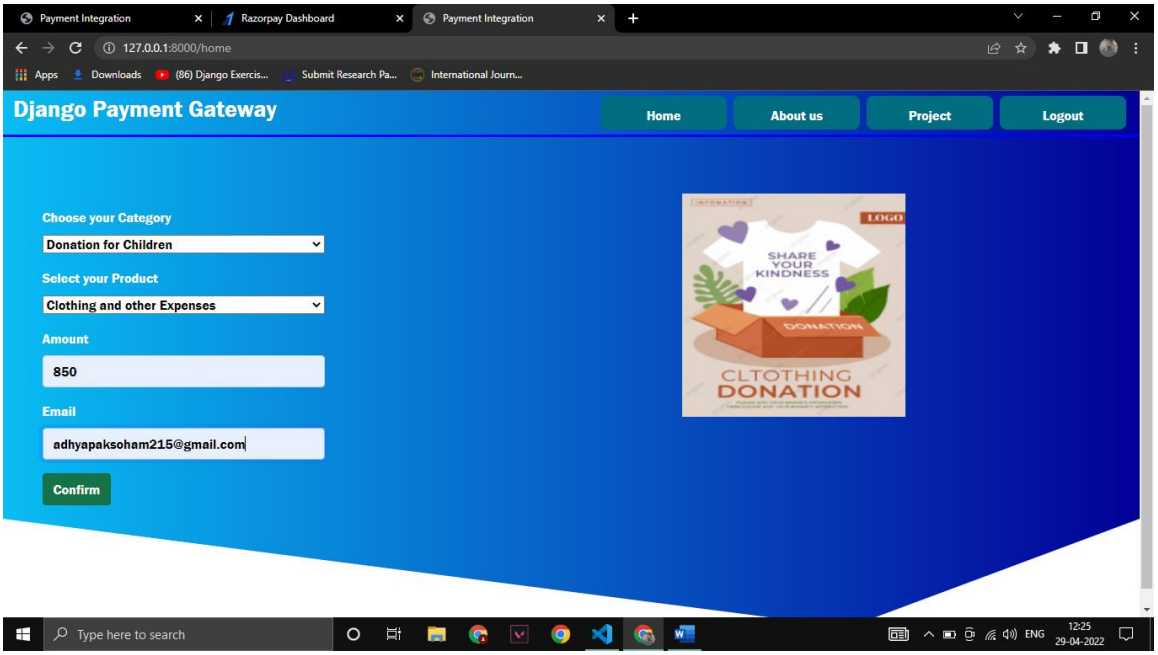
USER MANUAL

USER MANUAL

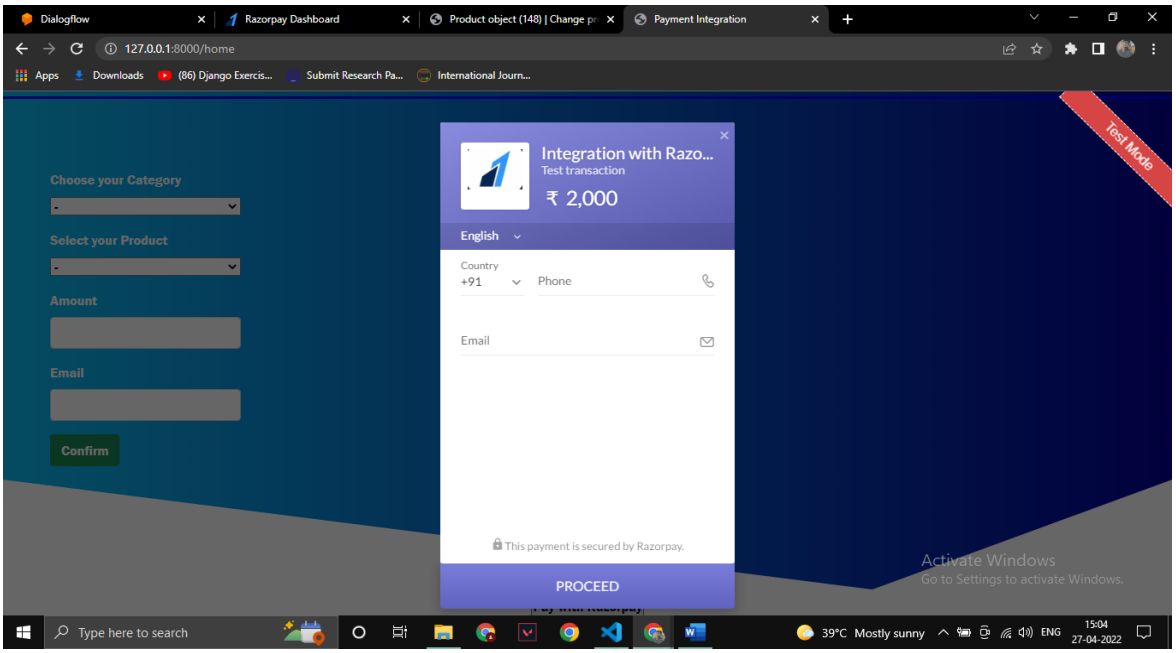
Step 1: Use credentials to login or create new account

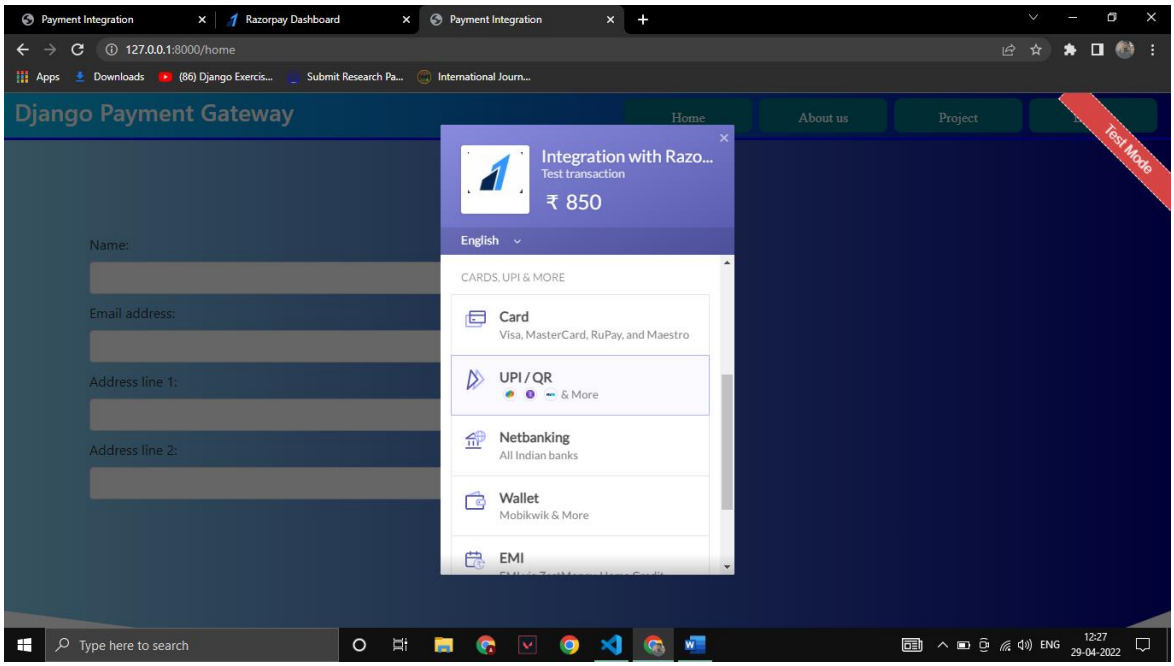


Step 2: Choose your donation and proceed for Checkout for payment.

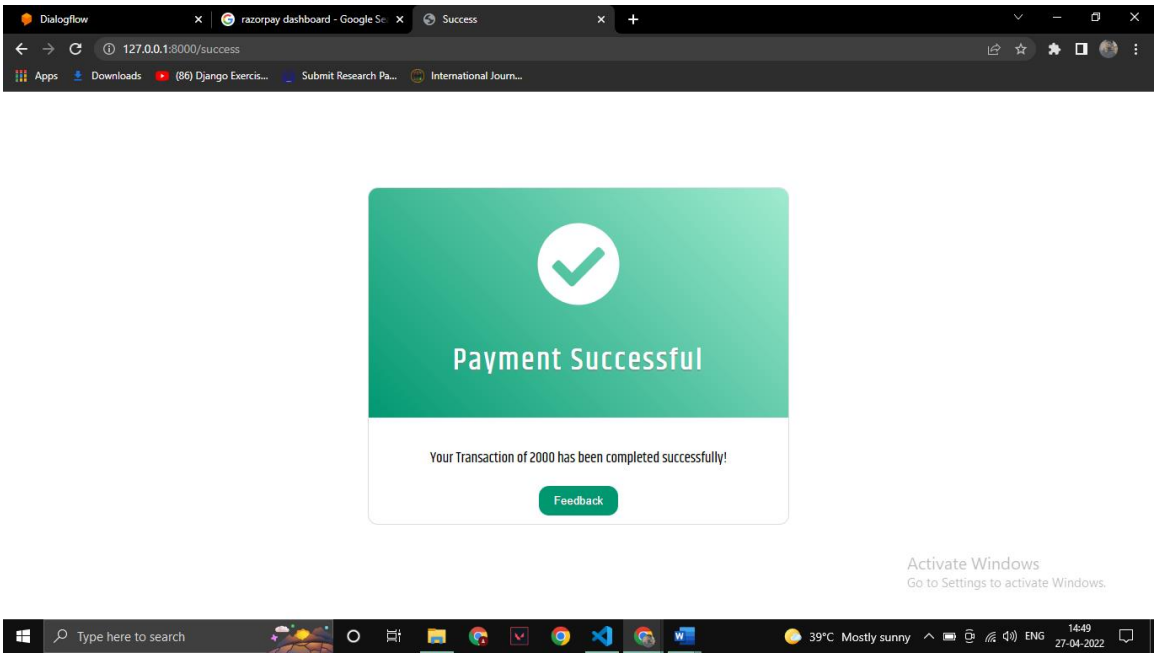


Step 3: Select your mode of payment.

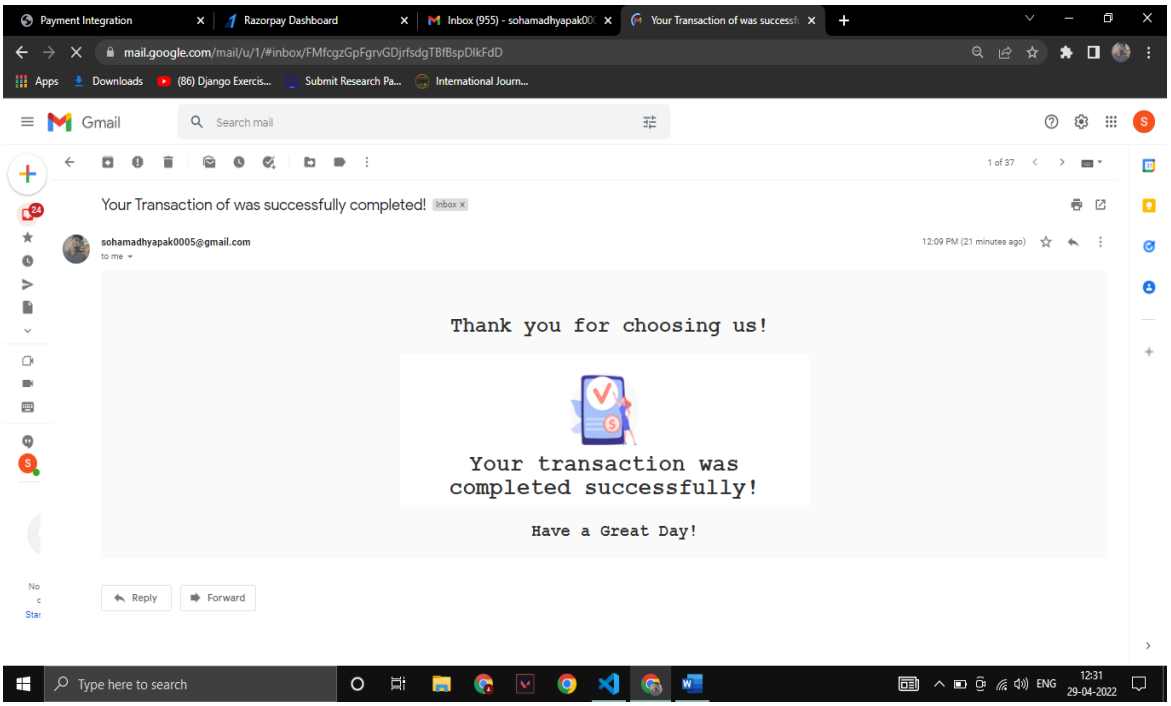




Step 4: complete your transaction and wait for payment status



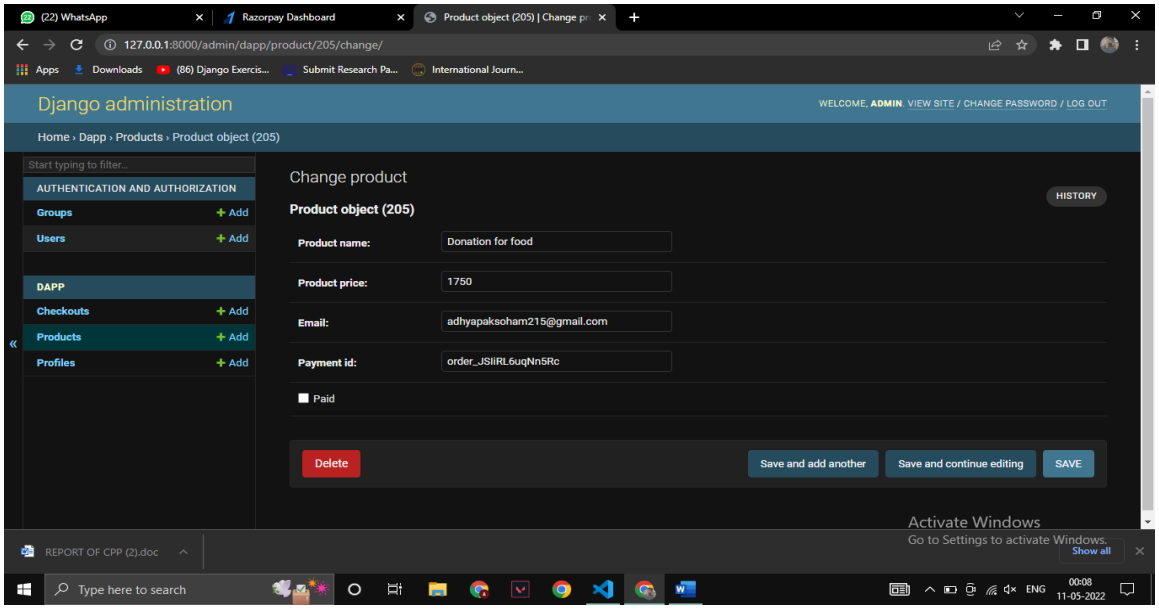
Step 5: Check your email for confirmation of payment

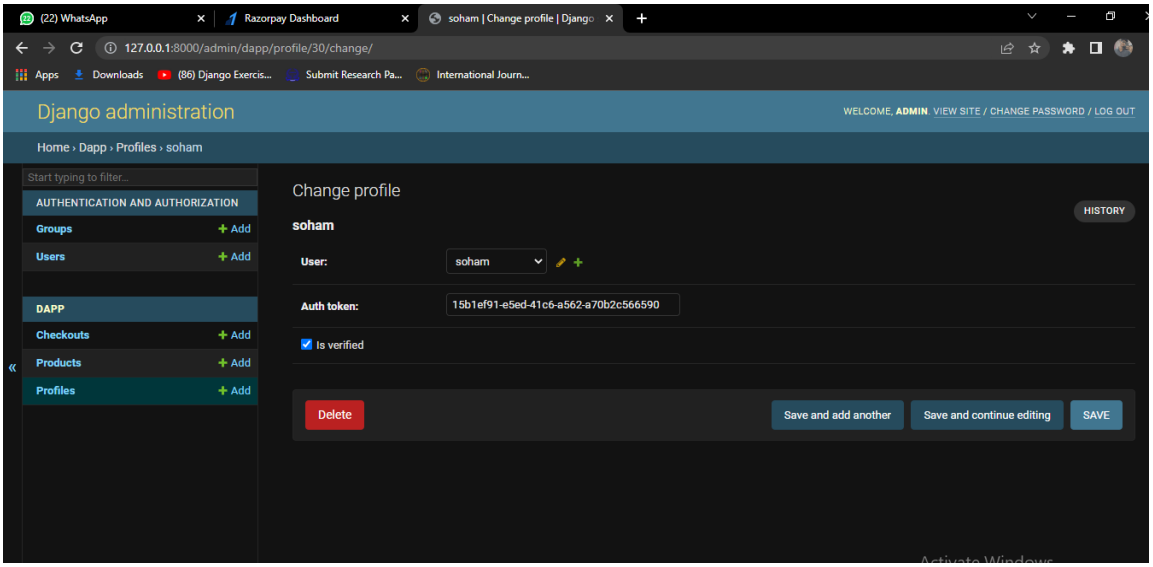


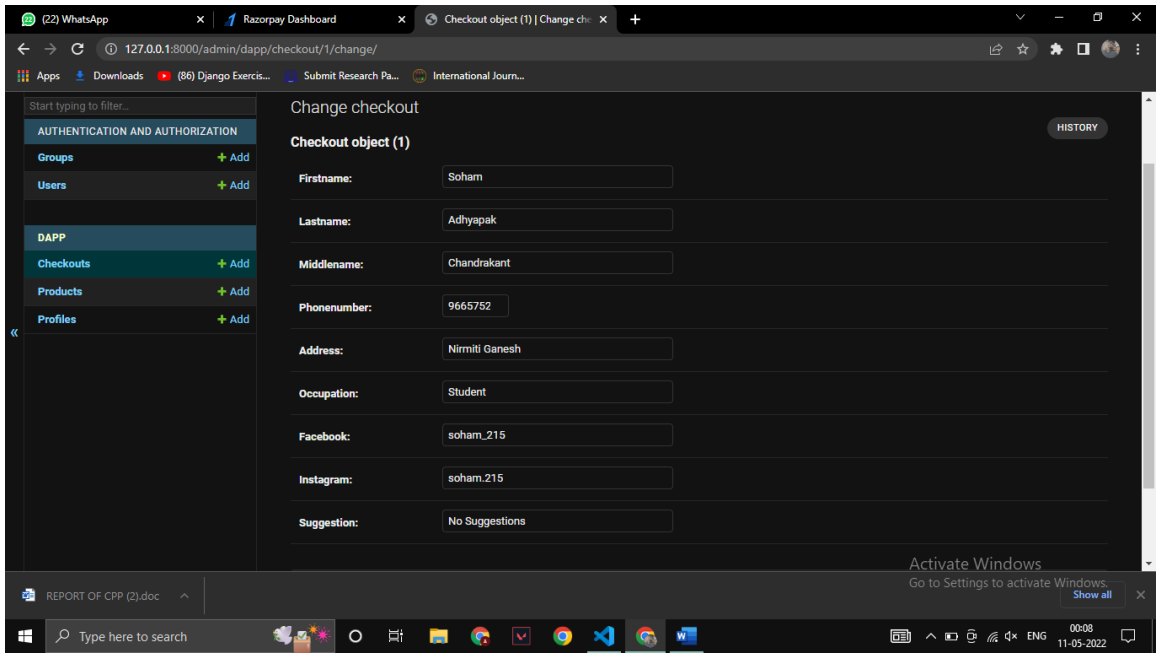
RESULT

➢ Functional Partitioning:

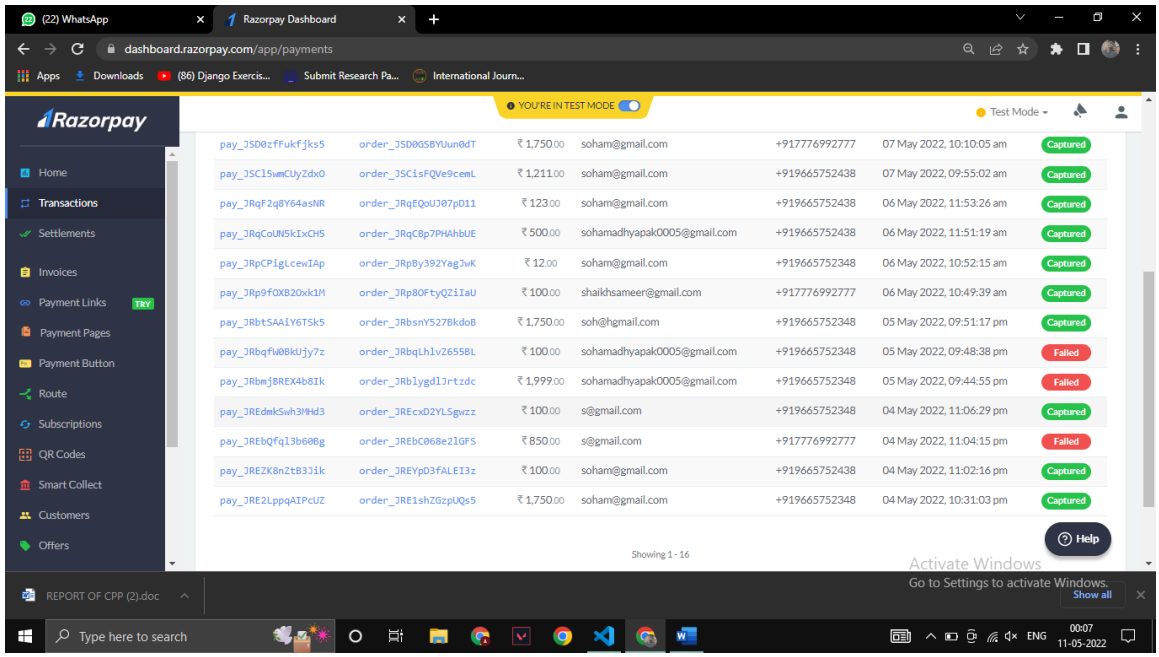
1. Database







2. Razorpay Dashboard



APPLICATION

APPLICATIONS

This project can be modified and used in following areas:

- ✓ E-Donation Systems.
- ✓ Reservation Portals.
- ✓ E-Commerce Website
- ✓ Third Party Agent Systems.
- ✓ Banking Systems.

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